



E-commerce Order Fulfillment Automation Backend

Saika Sarara · 230042159

Adiba Ahsan Raiyan · 230042161

Maliha Tasnim Khan · 230042127

The Problem



**Manual
Fulfillment**

= Costly Errors

during high-volume periods

Orders entered incorrectly — no validation layer

Stock mismatched — products oversold

Invoices skipped, no payment audit trail

Shipment updates delayed or missed entirely

No log to trace what happened to an order

Project Overview



What

A Java-based headless CLI backend that automates post-order fulfillment workflows for small-to-medium e-commerce businesses.



Who

Admin staff using a role-aware dashboard (ADMIN, MANAGER, SUPPORT) to manage orders, stock, payments, and reports.



Why

Eliminates manual errors, ensures stock accuracy, simulates payment, tracks shipment stages, and maintains a full audit trail.

Key Features



Order Validation

Rejects incomplete orders, invalid quantities, and malformed data before processing.



Inventory Mgmt.

Stock verification, reservation, restocking, low-stock alerts, and advanced filtering.



Payment Simulation

COD and MockCard approval/decline flow with payment outcome logging.



Shipping Workflow

Status progression: PACKED → SHIPPED → OUT_FOR_DELIVERY → DELIVERED.



Reports & Logs

Order summaries, revenue reports, receipt generation, and full audit logging.



Security & Roles

SHA-256 password hashing, login audit, and ADMIN / MANAGER / SUPPORT roles.

Tools & Technology

Java

Core language — OOP, modular class design

CLI (Console)

Command-line interface, headless application

JSON Files

Local persistence layer (no external DB)

SHA-256

Password hashing for secure authentication

Git & GitHub

Version control and team collaboration



Design Constraints

- `System.out.print()` for output
- `BufferedReader` for input & file read
- `FileWriter` for file writing
- `String.split()` for text parsing
- `MessageDigest` for SHA-256 hashing
- No external libraries — built from scratch

System Architecture

Entry Point	Main.java App init, login, startup/shutdown
Controller	Workflow.java Dashboard, order flow, reports, archive
Data Layer	DataPersistence.java JSON load/save, ID gen, totals
Services	PaymentService.java COD & MockCard payment simulation
Entities	Order • Product • Item • Log • Admin • Role Core domain models



JSON Files

- admins.json
- products.json
- orders.json
- logs.json
- login_audit.json
- stock_report.json
- report.json
- receipt files

Project Timeline — Gantt Chart



Team Contributions

Saika Sarara

230042159

*Workflow.java · Main.java ·
ManualInputReader.java*

Led workflow orchestration, admin dashboard, menu handling, order search & filter, reporting, archive operations, simulation flow, and application startup/shutdown.

Maliha Tasnim Khan

230042127

*DataPersistence.java ·
PaymentService.java*

JSON-based load/save, ID normalization & generation, total calculation, login audit writing, test-data support, and payment simulation (COD & MockCard).

Adiba Ahsan Raiyan

230042161

*Admin · Item · Log · Order · Product ·
Role*

Authentication & SHA-256 hashing, role-aware admin model, core entity classes, workflow log writing, and order timeline viewing.

All members collaborated on requirements, debugging, testing, documentation, and presentations.

Future Work

Web / GUI Frontend

Add a web dashboard or graphical interface for broader usability.

Relational Database

Replace JSON with SQL/NoSQL for scalability and concurrent access.

Customer Management

Order history analytics, richer invoicing, and customer profiles.

Real Payment Gateway

Integrate live APIs (SSLCommerz, bKash) instead of simulation.

Granular Permissions

Fine-grained role-based authorization and access control.

Automated Testing

Expand test coverage for robustness under production workloads.

Thank You



E-commerce Order Fulfillment Automation Backend

A Java CLI backend that automates post-order workflows — validation, inventory, payment, shipping, logging, and reporting.

github.com/saikasarara645-bot/E-commerce-Order-Fulfillment-Automation-Backend

Saika Sarara · Adiba Ahsan Raiyan · Maliha Tasnim Khan

Group 13